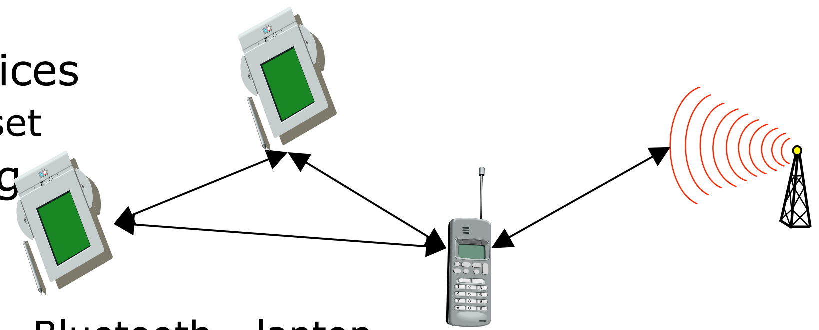
Bluetooth and Mobile IP

Bluetooth

- Consortium: Ericsson, Intel, IBM, Nokia, Toshiba…
- Scenarios:
  - connection of peripheral devices
    - loudspeaker, joystick, headset
  - support of ad-hoc networking
    - small devices, low-cost
  - bridging of networks
    - e.g., GSM via mobile phone - Bluetooth - laptop



- Simple, cheap, replacement of IrDA, low range, lower data rates, low-power
  - Worldwide operation: 2.4 GHz
  - Resistance to jamming and selective frequency fading:
    - FHSS over 79 channels (of 1MHz each), 1600hops/s
  - Coexistence of multiple piconets: like CDMA
  - Links: synchronous connections and asynchronous connectionless
  - Interoperability: protocol stack supporting TCP/IP, OBEX, SDP
  - Range: 10 meters, can be extended to 100 meters
- Documentation: over 1000 pages specification: www.bluetooth.com

Bluetooth Application Areas

- Data and voice access points
  - Real-time voice and data transmissions
- Cable replacement
  - Eliminates need for numerous cable attachments for connection
- Low cost < $5
- Ad hoc networking
  - Device with Bluetooth radio can establish connection with another when in range

Protocol Architecture

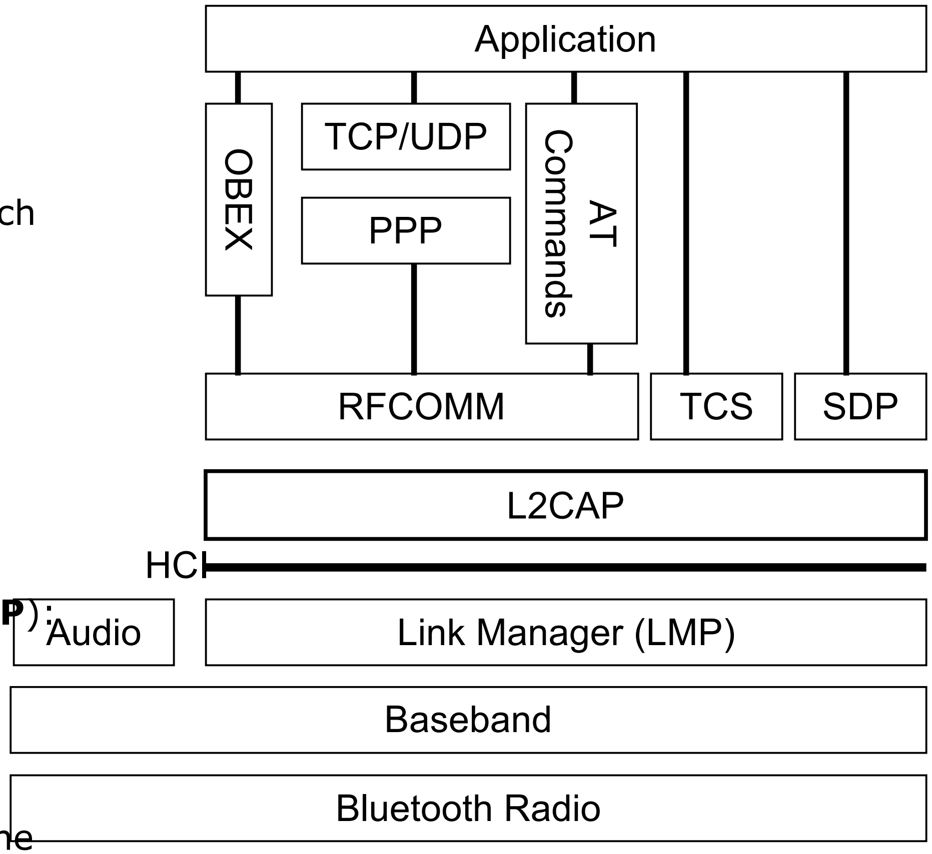
- Bluetooth is a layered protocol architecture
  - Core protocols
  - Cable replacement and telephony control protocols
  - Adopted protocols
- Core protocols
  - Radio
  - Baseband
  - Link manager protocol (LMP)
  - Logical link control and adaptation protocol (L2CAP)
  - Service discovery protocol (SDP)

Protocol Architecture

- Cable replacement protocol
  - RFCOMM
- Telephony control protocol
  - Telephony control specification – binary (TCS BIN)
- Adopted protocols
  - PPP
  - TCP/UDP/IP
  - OBEX
  - WAE/WAP

Protocol Architecture

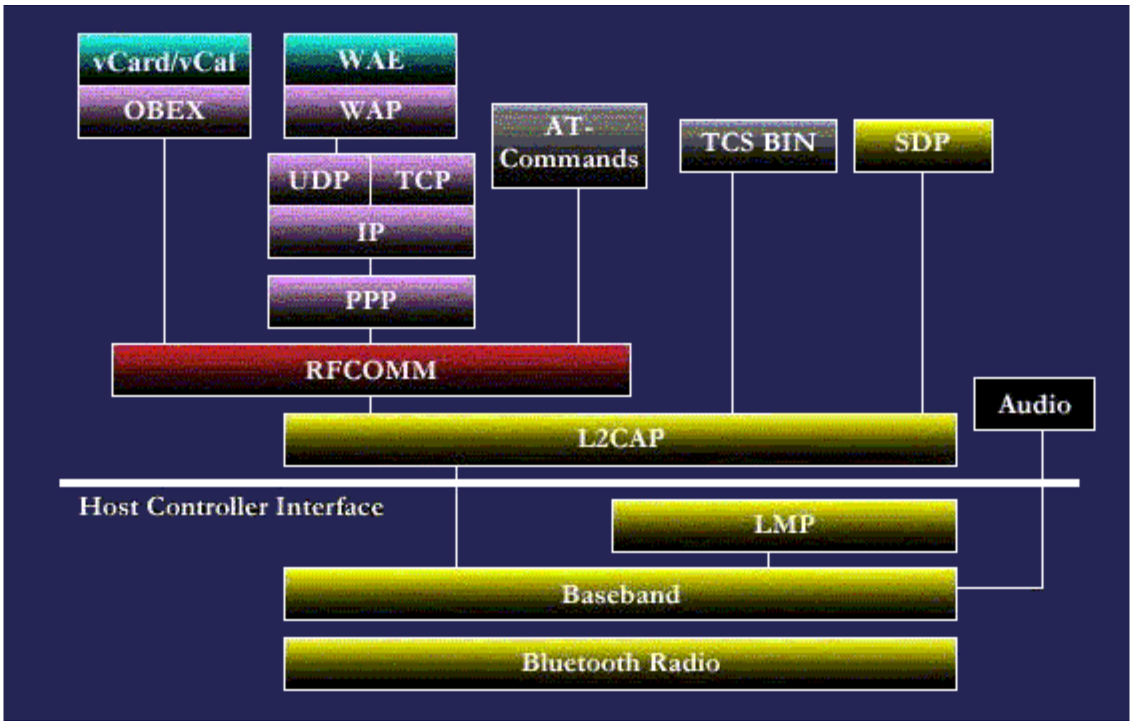
- BT **Radio** (2.4 GHZ Freq. Band):
- Modulation: Gaussian Frequency Shift Keying
- **Baseband**: FH-SS (79 carriers), CDMA (hopping sequence from the node MAC address)
- **Audio**: interfaces directly with the baseband. Each voice connection is over a 64Kbps SCO link. The voice coding scheme is the Continuous Variable Slope Delta (CVSD)
- Link Manager Protocol (**LMP**): link setup and control, authentication and encryption
- Host Controller Interface: provides a uniform method of access to the baseband, control registers, etc through USB, PCI, or UART
- Logical Link Control and Adaptation Layer (**L2CAP**): higher protocols multiplexing, packet segmentation/reassembly, QoS
- Service Discover Protocol (**SDP**): protocol of locating services provided by a Bluetooth device
- Telephony Control Specification (**TCS**): defines the call control signaling for the establishment of speech and data calls between Bluetooth devices
- **RFCOMM**: provides emulation of serial links (RS232). Upto 60 connections



OBEX: OBject EXchange (e.g., vCard)

What is Bluetooth? Bluetooth is the term used to describe the protocol of a short range (10 meter) frequency-hopping radio link between devices. These devices are then termed Bluetooth - enabled. Documentation on Bluetooth is split into two sections, the Bluetooth Specification and Bluetooth Profiles. The Specification describes how the technology works (i.e the Bluetooth protocol architecture), the Profiles describe how the technology is used (i.e how different parts of the specification can be used to fulfil a desired function for a Bluetooth device)

Bluetooth Specification Protocol Stack:



Bluetooth is the name given to a new technology using short-range radio links, intended to replace the cable(s) connecting portable and/or fixed electronic devices. Its key features are robustness, low complexity, low power and low cost. Designed to operate in noisy frequency environments, the Bluetooth radio uses a fast acknowledgement and frequency hopping scheme to make the link robust. Bluetooth radio modules operate in the unlicensed ISM band at 2.4GHz, and avoid interference from other signals by hopping to a new frequency after transmitting or receiving a packet.

Bluetooth Radio:

The Bluetooth Radio (layer) is the lowest defined layer of the Bluetooth specification. It defines the requirements of the Bluetooth transceiver device operating in the 2.4GHz ISM band.

1.1 Frequency Bands and Channel Arrangement

The Bluetooth radio accomplishes spectrum spreading by frequency hopping in 79 hops displaced by 1 MHz, starting at 2.402GHz and finishing at 2.480GHz. A guard band is used at the lower and upper band edge

1.2 Transmitter Characteristics

Power Classes: Each device is classified into 3 power classes, Power Class 1, 2 & 3.

Power Class 1: is designed for long range (~100m) devices, with a max output power of 20 dBm,

Power Class 2: for ordinary range devices (~10m) devices, with a max output power of 4 dBm,

Power Class 3: for short range devices (~10cm) devices, with a max output power of 0 dBm.

The Bluetooth radio interface is based on a nominal antenna power of 0dBm. Each device can optionally vary its transmitted power. Equipment with power control capability optimizes the output power in a link with LMP commands (Link Manager Protocol). It is done by measuring RSSI and reporting back if the power should be increased or decreased.

Modulation Characteristics: The Bluetooth radio module uses GFSK (Gaussian Frequency Shift Keying) where a binary one is represented by a positive frequency deviation and a binary zero by a negative frequency deviation.

Bluetooth Baseband

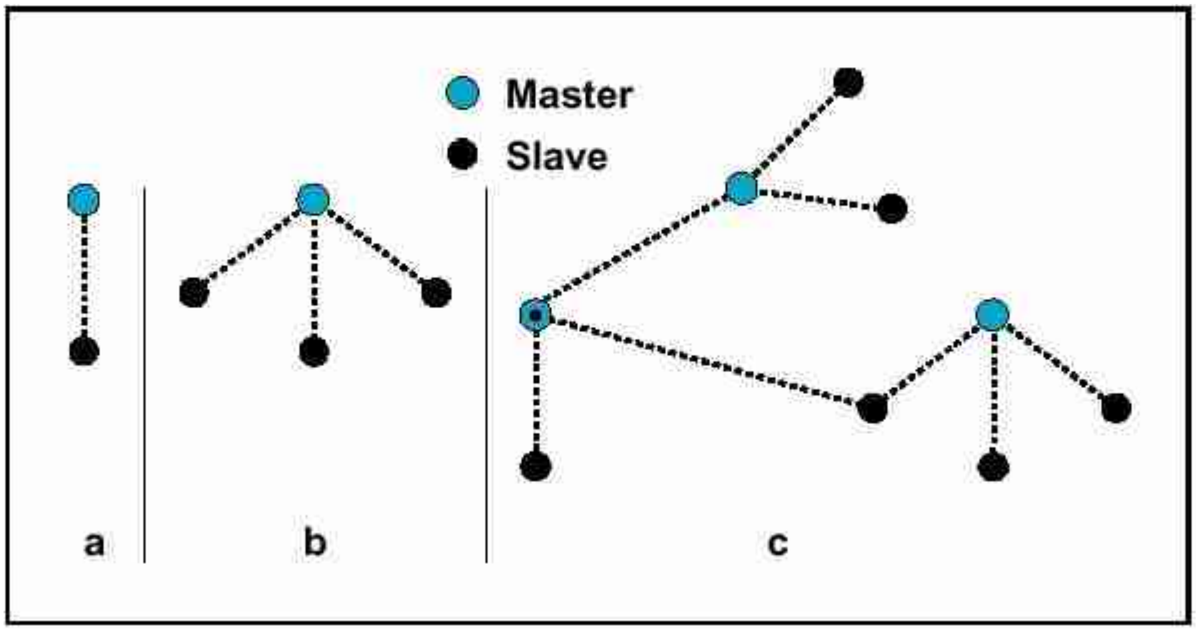
The Baseband is the physical layer of the Bluetooth. It manages physical channels and links apart from other services like error correction, data whitening, hop selection and Bluetooth security. The baseband protocol is implemented as a Link Controller, which works with the link manager for carrying out link level routines like link connection and power control. The baseband also manages asynchronous and synchronous links, handles packets and does paging and inquiry to access and inquire Bluetooth devices in the area. The baseband transceiver applies a time-division duplex (TDD) scheme (alternate transmit and receive). Therefore apart from different hopping frequency (frequency division), time is also slotted.

2.1 Physical Characteristics

2.1.1 Physical Channel

Bluetooth operates in the 2.4 GHz ISM band. In the US and Europe, a band of 83.5 MHz width is available; in this band, 79 RF channels spaced 1 MHz apart are defined. The channel is represented by a pseudo-random hopping sequence hopping through the 79 RF channels.

Two or more Bluetooth devices using the same channel form a piconet. There is one master and one or more slave(s) in each piconet. The hopping sequence is unique for the piconet and is determined by the Bluetooth device address (BD_ADDR) of the master; the phase in the hopping sequence is determined by the Bluetooth clock of the master. The channel is divided into time slots where each slot corresponds to an RF hop frequency. Consecutive hops correspond to different RF hop frequencies.



The channel is divided into time slots, each 625 us in length. The time slots are numbered according to the Bluetooth clock of the piconet master.

A TDD scheme is used where master and slave alternatively transmit. The master shall start its transmission in even-numbered time slots only, and the slave shall start its transmission in odd-numbered time slots only. The packet start shall be aligned with the slot start.

2.1.2 Physical Links

The Baseband handles two types of links: SCO (Synchronous Connection-Oriented) and ACL (Asynchronous Connection-Less) link. The SCO link is a symmetric point-to-point link between a master and a single slave in the piconet. The master maintains the SCO link by using reserved slots at regular intervals (circuit switched type). The SCO link mainly carries voice information. The master can support up to three simultaneous SCO links while slaves can support two or three SCO links. SCO packets are never retransmitted. SCO packets are used for 64 kB/s speech transmission.

The ACL link is a point-to-multipoint link between the master and all the slaves participating on the piconet. In the slots not reserved for the SCO links, the master can establish an ACL link on a per-slot basis to any slave, including the slave already engaged in an SCO link (packet switched type). Only a single ACL link can exist. For most ACL packets, packet retransmission is applied.

2.1.3 Logical Channels

Bluetooth has five logical channels which can be used to transfer different types of information. LC (Link Control Channel) and LM (Link Manager) channels are used in the link level while UA, UI and US channels are used to carry asynchronous, isosynchronous and synchronous user information.

2.3 Channel Control

2.3.1 Controller States

Bluetooth controller operates in two major states: Standby and Connection. There are seven substates which are used to add slaves or make connections in the piconet. These are page, page scan, inquiry, inquiry scan, master response, slave response and inquiry response .

The Standby state is the default low power state in the Bluetooth unit. Only the native clock is running and there is no interaction with any device whatsoever. In the Connection state, the master and slave can exchange packet, using the channel (master) access code and the master Bluetooth clock. The hopping scheme used is the channel hopping scheme.

2.3.2 Connection Setup (Inquiry/Paging)

Normally, a connection between two devices occur in the following fashion: If nothing is known about a remote device, both the inquiry(1) and page(2) procedures have to be followed. If some details are known about a remote device, only the paging procedure (2) is needed

Step 1:

The inquiry procedure enables a device to discover which devices are in range, and determine the addresses and clocks for the devices.

1.1:	The inquiry procedure involves a unit (the source) sending out inquiry packets (inquiry state) and then receiving the inquiry reply
1.2:	The unit that receives the inquiry packets (the destination), will hopefully be in the inquiry scan state to receive the inquiry packets.
1.3:	The destination will then enter the inquiry response state and send an inquiry reply to the source.

After the inquiry procedure has completed, a connection can be established using the paging procedure.

Step 2:

With the paging procedure, an actual connection can be established. The paging procedure typically follows the inquiry procedure. Only the Bluetooth device address is required to set up a connection. Knowledge about the clock (clock estimate) will accelerate the setup procedure. A unit that establishes a connection will carry out a page procedure and will automatically be the master of the connection. The procedure occurs as follows:

2.1:	A device (the source) pages another device (the destination).	Page state
2.2:	The destination receives the page.	Page Scan state

2.3:	The destination sends a reply to the source.	Slave Response state : (Step 1)
2.4:	The source sends an FHS packet to the destination.	Master Response state : (Step 1)
2.5:	The destination sends it's second reply to the source.	Slave Response state : (Step 2)
2.6:	The destination & source then switch to the source channel parameters.	Master Response state: Step 2 & Slave Response state: Step 3

The Connection state starts with a POLL packet sent by the master to verify that slave has switched to the master's timing and channel frequency hopping. The slave can respond with any type of packet.

2.3.3 Connection Modes

A Bluetooth device in the Connection state can be in any of the four following modes: Active, Hold, Sniff and Park mode.

Active Mode: In the active mode, the Bluetooth unit actively participates on the channel. The master schedules the transmission based on traffic demands to and from the different slaves. In addition, it supports regular transmissions to keep slaves synchronized to the channel. Active slaves listen in the master-to-slave slots for packets. If an active slave is not addressed, it may sleep until the next new master transmission.

Sniff Mode: Devices synchronized to a piconet can enter power-saving modes in which device activity is lowered. In the SNIFF mode, a slave device listens to the piconet at reduced rate, thus reducing its duty cycle. The SNIFF interval is programmable and depends on the application. It has the highest duty cycle (least power efficient) of all 3 power saving modes (sniff, hold & park).

Hold Mode: Devices synchronized to a piconet can enter power-saving modes in which device activity is lowered. The master unit can put slave units into HOLD mode, where only an internal timer is running. Slave units can also demand to be put into HOLD mode. Data transfer restarts instantly when units transition out of HOLD mode. It has an intermediate duty cycle (medium power efficient) of the 3 power saving modes (sniff, hold & park).

Park Mode: In the PARK mode, a device is still synchronized to the piconet but does not participate in the traffic. Parked devices have given up their MAC (AM_ADDR) address and occasional listen to the traffic of the master to re-synchronize and check on broadcast messages. It has the lowest duty cycle (power efficiency) of all 3 power saving modes (sniff, hold & park).

2.3.4 Scatternet

Multiple piconets may cover the same area. Since each piconet has a different master, the piconets hop independently, each with their own channel hopping sequence and phase as determined by the respective master.

2.4 Other Baseband Functions

2.4.1 Error Correction

There are three kinds of error correction schemes used in the baseband protocol: 1/3 rate FEC, 2/3 rate FEC and ARQ scheme.

2.4.2 Flow Control

The Baseband protocol recommends using FIFO queues in ACL and SCO links for transmission and receive. The Link Manager fills these queues and link controller empties the queues automatically.

2.4.3 Synchronization

The Bluetooth transceiver uses a time-division duplex (TDD) scheme, meaning that it alternately transmits and receives in a synchronous manner.

The piconet is synchronized by the system clock of the master. To transmit on the piconet channel 3 pieces of information are needed. The (channel) hopping sequence, the phase of the sequence, and the CAC to place on the packets

1	Channel Hopping Sequence	The Bluetooth Device Address (BD_ADDR) of the master is used to derive this frequency hopping sequence.
2	Phase	The system clock of the master determines the phase in the hopping sequence.
3	Channel Access Code	This is derived from the Bluetooth Device Address (BD_ADDR) of the master.

The slaves adapt their native clocks with a timing offset in order to match the master clock, giving then an estimated clock value. The offset is zero for the master as it's native clock is the master clock.

2.4.4 Bluetooth Security

At the link layer, security is maintained by authentication of the peers and encryption of the information. For this basic security we need a public address which is unique for each device (BD_ADDR), two secret keys (authentication keys and encryption key) and a random number generator. First a device does the authentication by issuing a challenge and the other device has to then send a response to that challenge which is based on the challenge, it's BD_ADDR and a link key shared between them. After authentication, encryption may be used to communicate.

Link Manager Protocol (LMP)

The Link Manager carries out link setup, authentication, link configuration and other protocols. It discovers other remote LM's and communicates with them via the Link Manager Protocol (LMP). To perform its service provider role, the LM uses the services of the underlying Link Controller (LC).

The Link Manager Protocol essentially consists of a number of PDU (protocol Data Units), which are sent from one device to another, determined by the AM_ADDR in the packet header. LM PDUs are always sent as single-slot packets and the payload header is therefore one byte.

DM1 packets are used to transport LM PDUs except if an SCO link is present using HV1 packets and length of content is less than 9 bytes. In this case DV packets are used.

The following is a brief list of the types of PDU's available and their function/operation. These PDU are either Mandatory : M (must be supported), or Optional : O (optionally supported)

Logical Link Control and Adaptation Protocol

The Logical Link Control and Adaptation Layer Protocol (L2CAP) is layered over the Baseband Protocol and resides in the data link layer. L2CAP provides connection-oriented and connectionless data services to upper layer protocols with protocol multiplexing capability, segmentation and reassembly operation, and group abstractions. L2CAP permits higher level protocols and applications to transmit and receive L2CAP data packets up to 64 kilobytes in length.

Two link types are supported for the Baseband layer : Synchronous Connection-Oriented (SCO) links and Asynchronous Connection-Less (ACL) links. SCO links support real-time voice traffic using reserved bandwidth. ACL links support best effort traffic. The L2CAP Specification is defined for only ACL links and no support for SCO links is planned.

Slides:
http://www.ccs.neu.edu/home/rraj/Courses/6710/S10/Lectures/BluetoothMobileIP.pdf

Frequency-hopping spread spectrum (**FHSS**) is a method of transmitting radio signals by rapidly changing the carrier frequency among many distinct frequencies occupying a large spectral band. The changes are controlled by a code known to both transmitter and receiver. FHSS is used to avoid interference, to prevent eavesdropping, and to enable code-division multiple access (CDMA) communications.

The available frequency band is divided into smaller sub-bands. Signals rapidly change ("hop") their carrier frequencies among the center frequencies of these sub-bands in a predetermined order. Interference at a specific frequency will only affect the signal during a short interval.[1]

FHSS offers four main advantages over a fixed-frequency transmission:

1. FHSS signals are highly resistant to narrowband interference because the signal hops to a different frequency band.
2. Signals are difficult to intercept if the frequency-hopping pattern is not known.
3. Jamming is also difficult if the pattern is unknown; a malicious individual may only jam the signal for a single hopping period if the spreading sequence is unknown.
4. FHSS transmissions can share a frequency band with many types of conventional transmissions with minimal mutual interference. FHSS signals add minimal interference to narrowband communications, and vice versa.

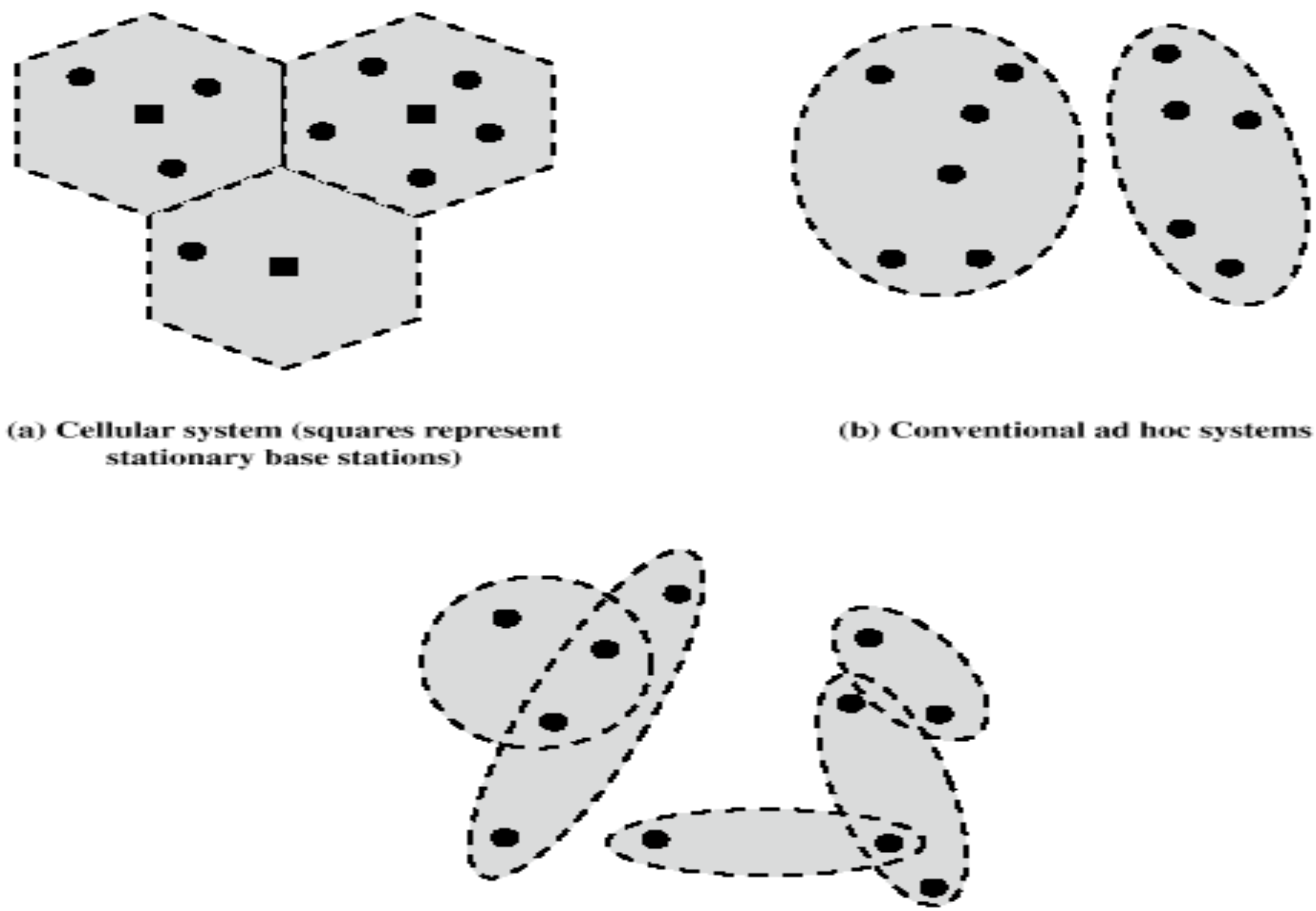
Usage Models

- File transfer
- Internet bridge
- LAN access
- Synchronization
- Three-in-one phone
- Headset

Piconets and Scatternets

- Piconet
  - Basic unit of Bluetooth networking
  - Master and one to seven slave devices
  - Master determines channel and phase
- Scatternet
  - Device in one piconet may exist as master or slave in another piconet
  - Allows many devices to share same area
  - Makes efficient use of bandwidth

Wireless Network Configurations



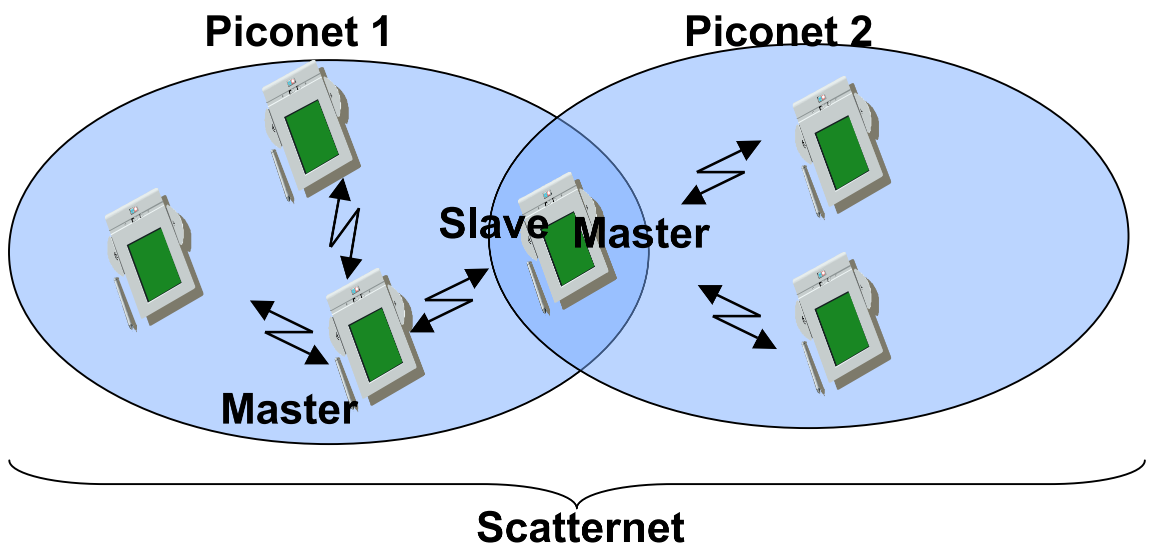
(a) Cellular system (squares represent stationary base stations)

(b) Conventional ad hoc systems

(c) Scatternets

Figure 15.5 Wireless Network Configurations

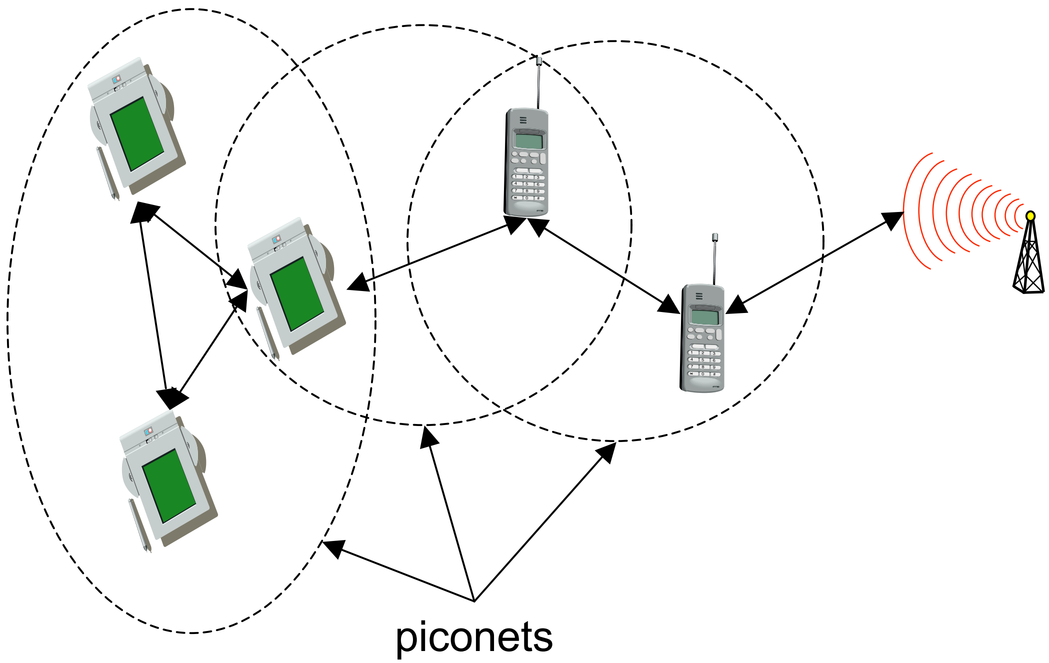
Network Topology



- Piconet = set of Bluetooth nodes synchronized to a master node
  - The piconet hopping sequence is derived from the master MAC address (BD_ADDR IEEE802 48 bits compatible address)
- Scatternet = set of piconet
- Master-Slaves can switch roles
- A node can only be master of one piconet. Why?

Scatternets

- Each piconet has one master and up to 7 slaves
- Master determines hopping sequence, slaves have to synchronize
- Participation in a piconet = synchronization to hopping sequence
- Communication between piconets = devices jumping back and forth between the piconets



Radio Specification

- Classes of transmitters
  - Class 1: Outputs 100 mW for maximum range
    - Power control mandatory
    - Provides greatest distance
  - Class 2: Outputs 2.4 mW at maximum
    - Power control optional
  - Class 3: Nominal output is 1 mW
    - Lowest power
- Frequency Hopping in Bluetooth
  - Provides resistance to interference and multipath effects
  - Provides a form of multiple access among co-located devices in different piconets

Frequency Hopping

- Total bandwidth divided into 1MHz physical channels
- FH occurs by jumping from one channel to another in pseudorandom sequence
- Hopping sequence shared with all devices on piconet
- Piconet access:
  - Bluetooth devices use time division duplex (TDD)
  - Access technique is TDMA
  - FH-TDD-TDMA

Frequency Hopping

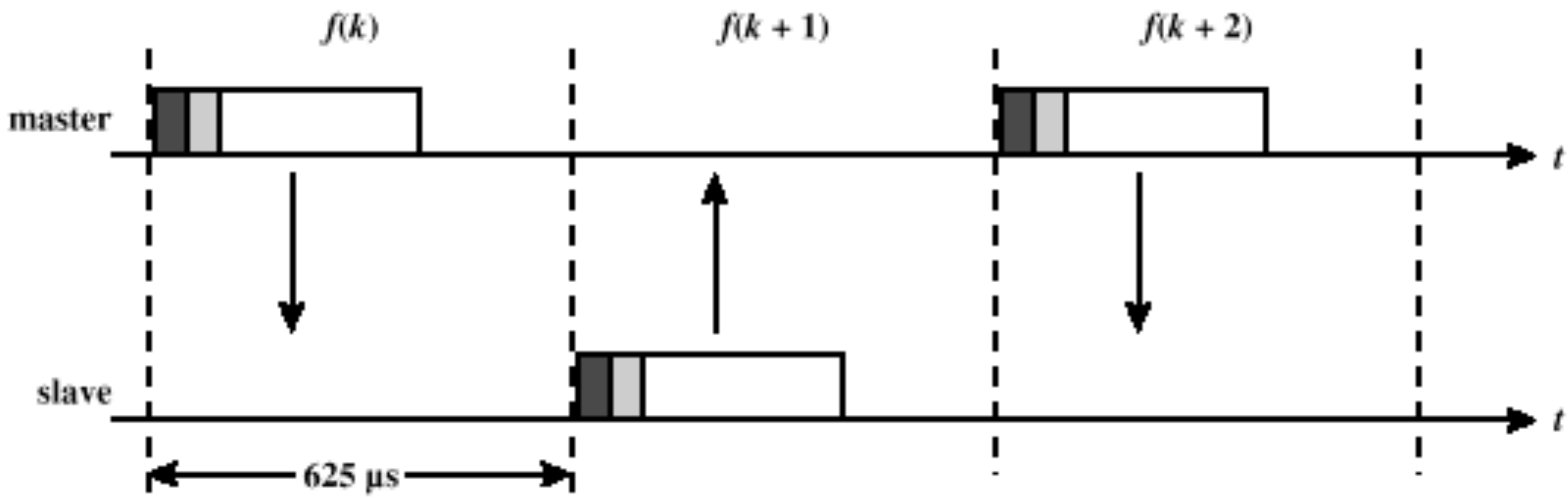


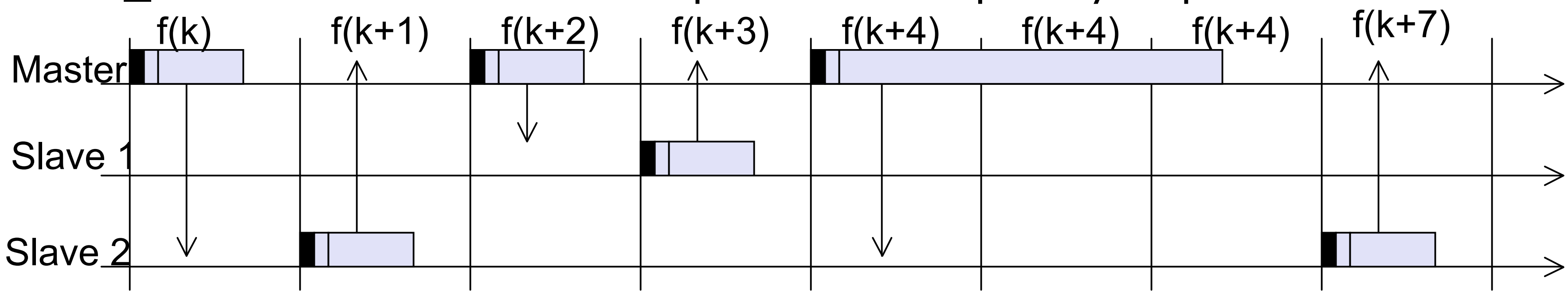
Figure 15.6 Frequency-Hop Time-Division Duplex

Physical Links

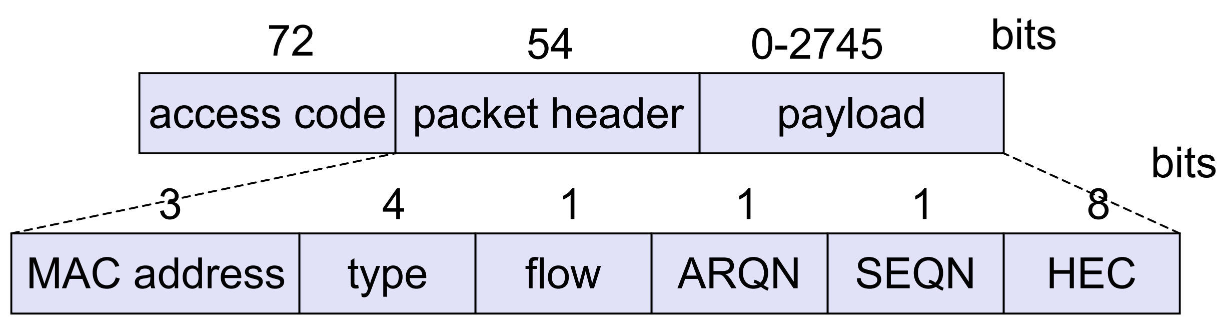
- Synchronous connection oriented (SCO)
  - Allocates fixed bandwidth between point-to-point connection of master and slave
  - Master maintains link using reserved slots
  - Master can support three simultaneous links
- Asynchronous connectionless (ACL)
  - Point-to-multipoint link between master and all slaves
  - Only single ACL link can exist

Bluetooth Piconet MAC

- Each node has a Bluetooth Device Address (BD_ADDR). The master BD_ADDR determines the sequence of frequency hops



- Types of connections:
  - Synchronous Connection-Oriented link (**SCO**) (symmetrical, circuit switched, point-to-point)
  - Asynchronous Connectionless Link (**ACL**): (packet switched, point-to-multipoint, master-polls)
- Packet Format:
  - o Access code: synchronization, when piconet active derived from master
  - o Packet header (for ACL): 1/3-FEC, MAC address (1 master, 7 slaves), link type, alternating bit ARQ/SEQ, checksum



Types of Access Codes

- Channel access code (CAC) – identifies a piconet
- Device access code (DAC) – used for paging and subsequent responses
- Inquiry access code (IAC) – used for inquiry purposes
- Preamble+sync+trailer

Packet Header Fields

- AM_ADDR – contains “active mode” address of one of the slaves
- Type – identifies type of packet
  - ACL: Data Medium (DM) or Data High (DH), with different slot lengths (DM1, DM3, DM5, DH1, DH3, DH5)
  - SCO: Data Voice (DV) and High-quality voice (HV)
- Flow – 1-bit flow control
- ARQN – 1-bit acknowledgment
- SEQN – 1-bit sequential numbering schemes
- Header error control (HEC) – 8-bit error detection code

Payload Format

- Payload header
  - o L_CH field – identifies logical channel
  - o Flow field – used to control flow at L2CAP level
  - o Length field – number of bytes of data
- Payload body – contains user data
- CRC – 16-bit CRC code

Error Correction Schemes

- 1/3 rate FEC (forward error correction)
  - Used on 18-bit packet header, voice field in HV1 packet
- 2/3 rate FEC
  - Used in DM packets, data fields of DV packet, FHS packet and HV2 packet
- ARQ
  - Used with DM and DH packets

ARQ Scheme Elements

- Error detection – destination detects errors, discards packets
- Positive acknowledgment – destination returns positive acknowledgment
- Retransmission after timeout – source retransmits if packet unacknowledged
- Negative acknowledgment and retransmission – destination returns negative acknowledgement for packets with errors, source retransmits
- Bluetooth uses ACKs, NAKs, and a form of stop-and-wait ARQ

Types of packets

- SCO packets: Do not have a CRC (except for the data part of DV) and are never retransmitted. Intended for High-quality Voice (HV).

Type	Payload (bytes)	FEC	CRC	max-rate kbps
HV1	10	1/3	No	64
HV2	20	2/3	No	64
HV3	30	No	No	64
DV	10+(1-10)D	2/3D	Yes D	64+57.6D

- ACL packets: Data Medium-rate (DM) and Data High-rate (DH)

Type	Payload (bytes)	FEC	CR C	Sym. max-rate	Asymm. max-rate (DL/UL)
DM1	0-17	2/3	Yes	108.8	108.8/108.9
DM3	0-121	2/3	Yes	258.1	387.2/54.4
DM5	0-224	2/3	Yes	286.7	477.8/36.3
DH1	0-27	No	Yes	172.8	172.8/172.8
DH3	0-183	No	Yes	390.4	585.6/86.4
DH5	0-339	No	Yes	433.9	723.2/185.6

Channel Control

- Major states
  - Standby – default state
  - Connection – device connected
- Interim substates for adding new slaves
  - Page – device issued a page (used by master)
  - Page scan – device is listening for a page
  - Master response – master receives a page response from slave
  - Slave response – slave responds to a page from master
  - Inquiry – device has issued an inquiry for identity of devices within range
  - Inquiry scan – device is listening for an inquiry
  - Inquiry response – device receives an inquiry response

Inquiry Procedure

- Potential master identifies devices in range that wish to participate
  - Transmits ID packet with inquiry access code (IAC)
  - Occurs in Inquiry state
- Device receives inquiry
  - Enter Inquiry Response state
  - Returns FHS (Frequency Hop Synchronization) packet with address and timing information
  - Moves to page scan state

Inquiry Procedure Details

- Goal: aims at discovering other neighboring devices
- Inquiring node:
  - Sends an inquiry message (packet with only the access code: General Inquiry Access Code: GIAC or Dedicated IAC: DIAC). This message is sent over a subset of all possible frequencies.
  - The inquiry frequencies are divided into two hopping sets of 16 frequencies each.
  - In inquiry state the node will send up to $N_{INQUIRY}$ sequences on one set of 16 frequencies before switching to the other set of 16 frequencies. Upto 3 switches can be executed. Thus the inquiry may last upto 10.24 seconds.
- To be discovered node:
  - Enters an inquiry_scan mode
  - When hearing the inquiry_message (and after a backoff procedure) enter an inquiry_response mode: send a Frequency Hop Sync (FHS) packet (BD_ADDR, native clock)
- After discovering the neighbors and collecting information on their address and clock, the inquiring node can start a page routine to setup a piconet

Page Procedure

- Master uses devices address to calculate a page frequency-hopping sequence
- Master pages with ID packet and device access code (DAC) of specific slave
- Slave responds with DAC ID packet
- Master responds with its FHS packet
- Slave confirms receipt with DAC ID
- Slaves moves to Connection state

Page Procedure Details

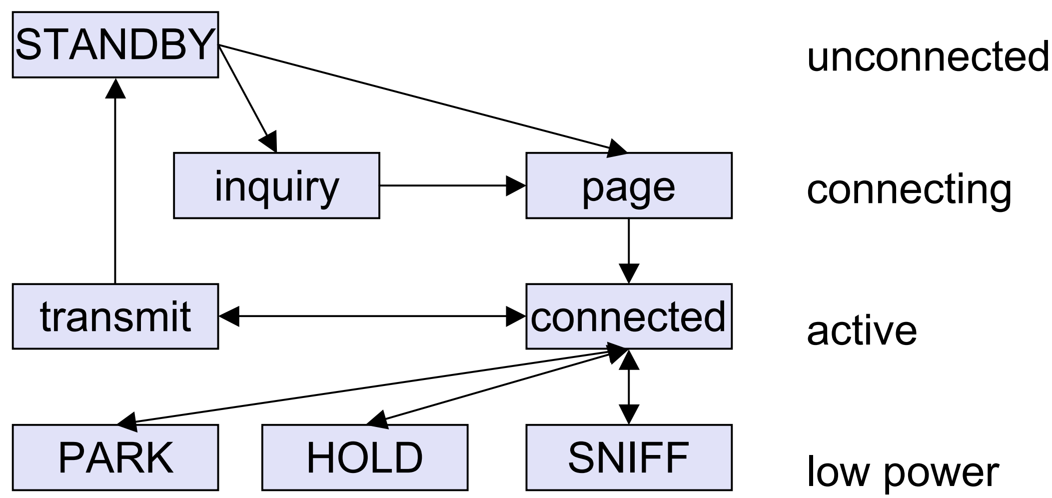
- Goal: e.g., setup a piconet after an inquiry
- Paging node (master):
  - Sends a page message (i.e., packet with only Device Access Code of paged node) over 32 frequency hops (from DAC and split into 2*16 freq.)
  - Repeated until a response is received
  - When a response is received send a FHS message to allow the paged node to synchronize
- Paged node (slave):
  - Listens on its hopping sequence
  - When receiving a page message, send a page_response and wait for the FHS of the pager

Slave Connection State Modes

- Active – participates in piconet
  - Listens, transmits and receives packets
- Sniff – only listens on specified slots
- Hold – does not support ACL packets
  - Reduced power status
  - May still participate in SCO exchanges
- Park – does not participate on piconet
  - Still retained as part of piconet

States of a Bluetooth Device

ACTIVE (connected/transmit): the device is uniquely identified by a 3bits AM_ADDR and is fully participating
SNIFF state: participates in the piconet only within the SNIFF interval
HOLD state: keeps only the SCO links
PARK state (low-power): releases AM_ADDR but stays synchronized with master



BT device addressing:

- BD_ADDR (48 bits)
- AM_ADDR (3bits): ACTIVE, HOLD, or SNIFF
- PM_ADDR (8 bits): PARK Mode address (exchanged with the AM_ADDR when entering PARK mode)
- AR_ADDR (8 bits): not unique used to come back from PARK to ACTIVE state

Bluetooth Audio

- Voice encoding schemes:
  - Pulse code modulation (PCM)
  - Continuously variable slope delta (CVSD) modulation
- Choice of scheme made by link manager
  - Negotiates most appropriate scheme for application

Bluetooth Link Security

- Elements:
  - Authentication – verify claimed identity
  - Encryption – privacy
  - Key management and usage
- Security algorithm parameters:
  - Unit address
  - Secret authentication key (128 bits key)
  - Secret privacy key (4-128 bits secret key)
  - Random number

Link Management

- Manages master-slave radio link
- Security Service: authentication, encryption, and key distribution
- Clock synchronization
- Exchange station capability information
- Mode management:
  - switch master/slave role
  - change hold, sniff, park modes
  - QoS

L2CAP

- Provides a link-layer protocol between entities with a number of services
- Relies on lower layer for flow and error control
- Makes use of ACL links, does not support SCO links
- Provides two alternative services to upper-layer protocols
  - Connectionless service
  - Connection-oriented service: A QoS flow specification is assigned in each direction
- Exchange of signaling messages to establish and configure connection parameters

Mobile IP

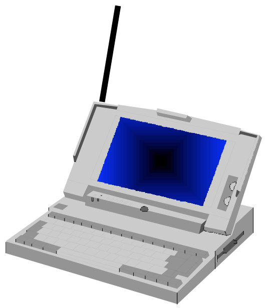
Motivation for Mobile IP

- Routing
  - based on IP destination address, network prefix (e.g. 129.13.42) determines physical subnet
  - change of physical subnet implies change of IP address to have a topological correct address (standard IP) or needs special entries in the routing tables
- Specific routes to end-systems?
  - change of all routing table entries to forward packets to the right destination
  - does not scale with the number of mobile hosts and frequent changes in the location, security problems
- Changing the IP-address?
  - adjust the host IP address depending on the current location
  - almost impossible to find a mobile system, DNS updates take too much time
  - TCP connections break, security problems

Mobile IP Requirements

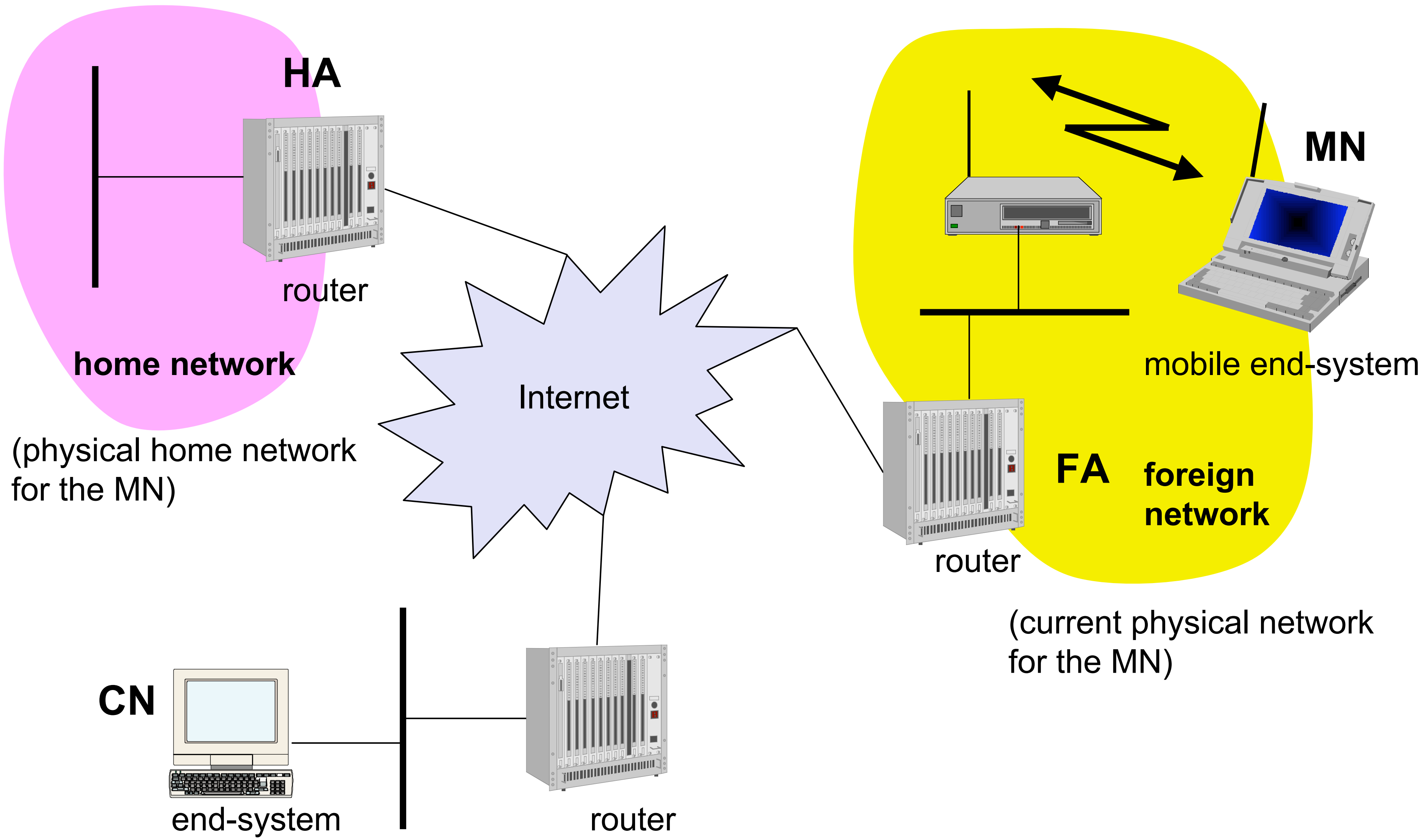
- Transparency
  - mobile end-systems keep their IP address
  - continuation of communication after interruption of link possible
  - point of connection to the fixed network can be changed
- Compatibility
  - support of the same layer 2 protocols as IP
  - no changes to current end-systems and routers required
  - mobile end-systems can communicate with fixed systems
- Security
  - authentication of all registration messages
- Efficiency and scalability
  - only little additional messages to the mobile system required (connection typically via a low bandwidth radio link)
  - world-wide support of a large number of mobile systems in the whole Internet

Terminology

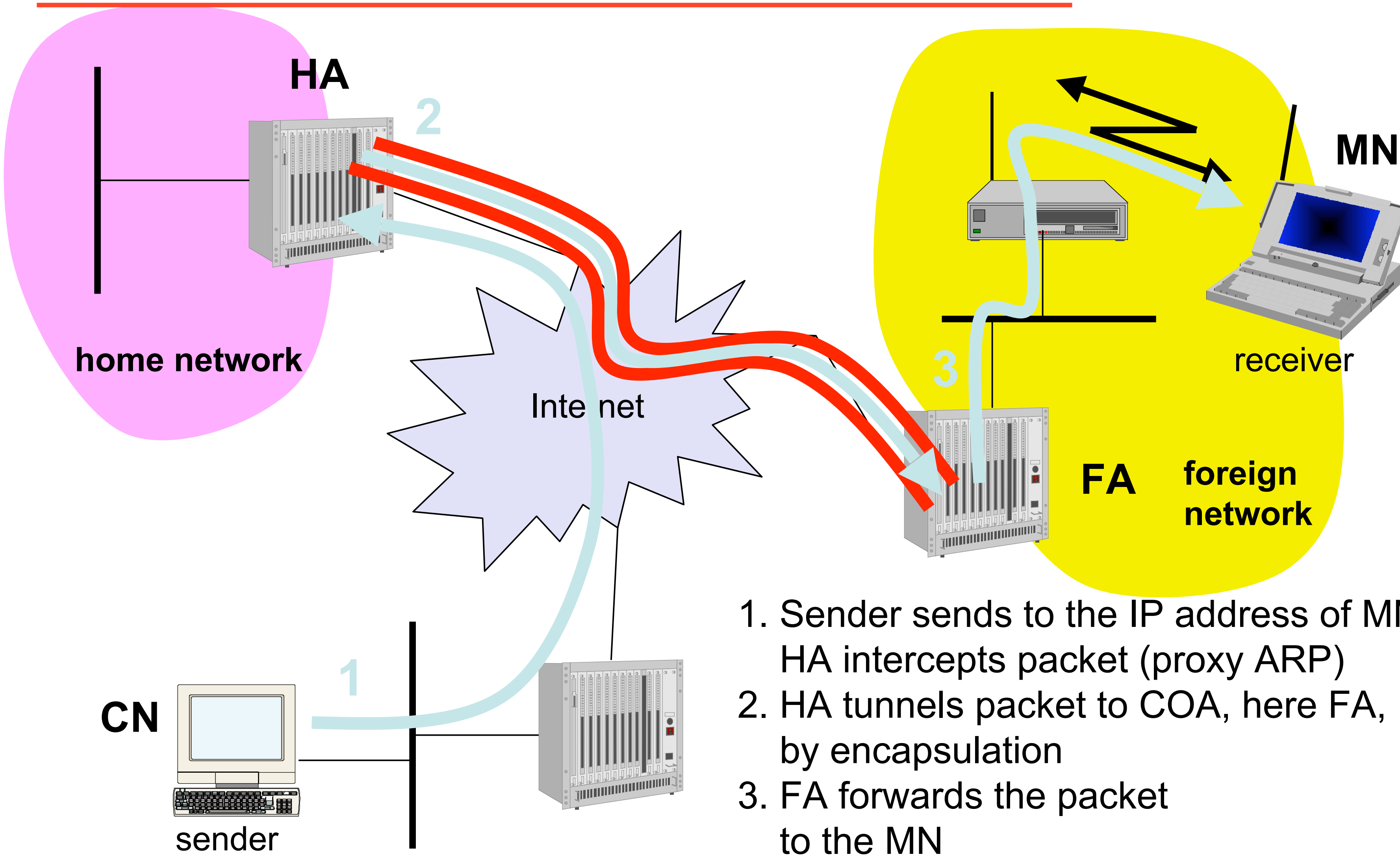


- Mobile Node (MN)
  - system (node) that can change the point of connection to the network without changing its IP address
- Home Agent (HA)
  - system in the home network of the MN, typically a router
  - registers the location of the MN, tunnels IP datagrams to the COA
- Foreign Agent (FA)
  - system in the current foreign network of the MN, typically a router
  - forwards the tunneled datagrams to the MN, typically also the default router for the MN
- Care-of Address (COA)
  - address of the current tunnel end-point for the MN (at FA or MN)
  - actual location of the MN from an IP point of view
  - can be chosen, e.g., via DHCP
- Correspondent Node (CN)
  - communication partner

Example network

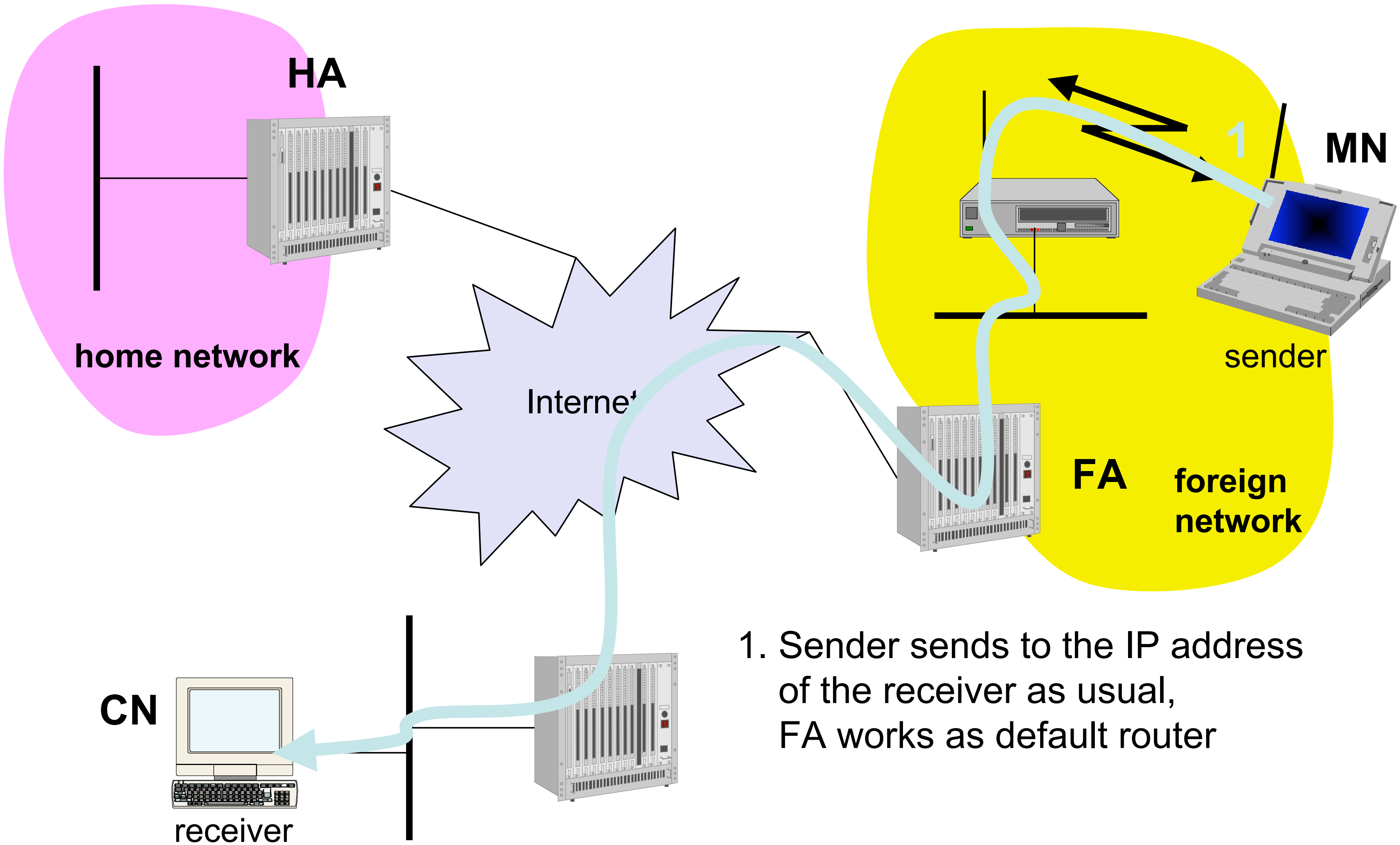


Data transfer to the mobile



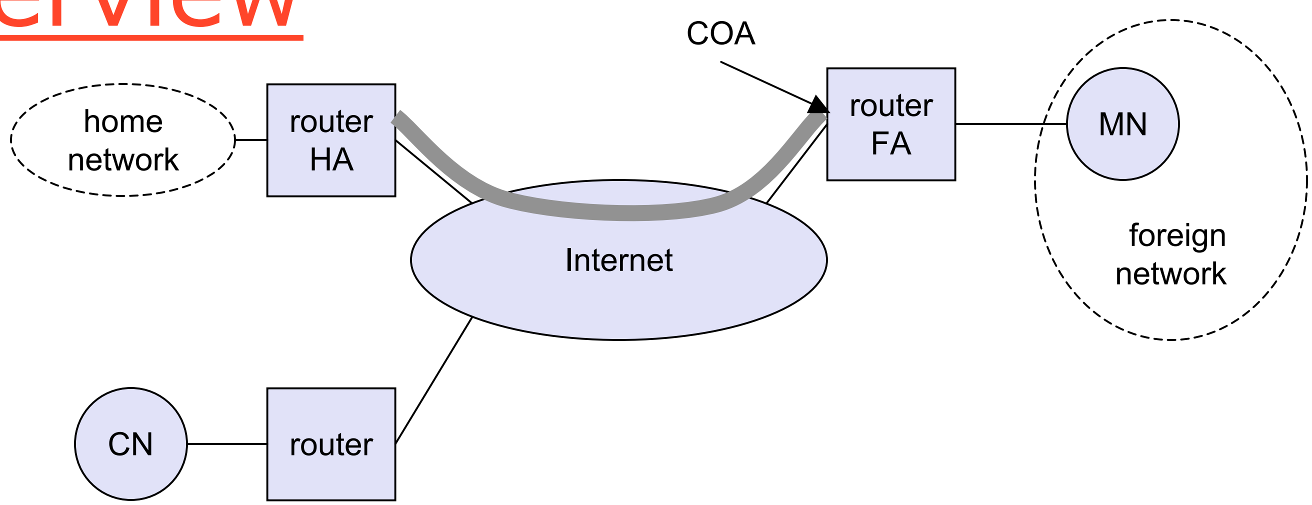
1. Sender sends to the IP address of MN, HA intercepts packet (proxy ARP)
2. HA tunnels packet to COA, here FA, by encapsulation
3. FA forwards the packet to the MN

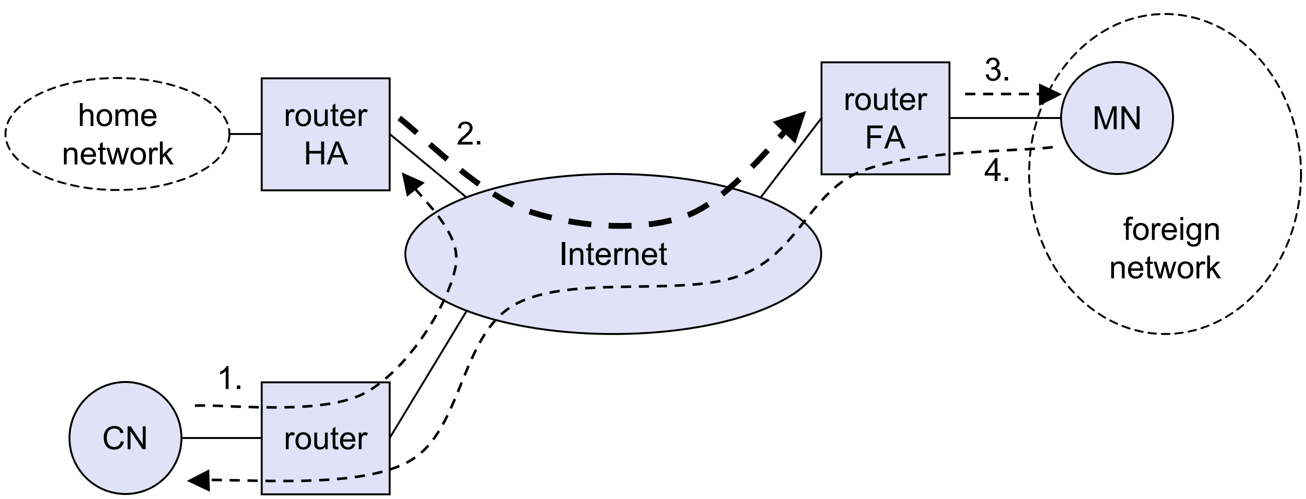
Data transfer from the mobile



1. Sender sends to the IP address of the receiver as usual, FA works as default router

Overview


COA
home network
router HA
router FA
MN
Internet
foreign network
CN
router


home network
router HA
2.
router FA
3.
MN
4.
Internet
foreign network
1.
CN
router

Network integration

- Agent Advertisement
  - HA and FA periodically send advertisement messages into their physical subnets
  - MN listens to these messages and detects, if it is in the home or a foreign network (standard case for home network)
  - MN reads a COA from the FA advertisement messages
- Registration (always limited lifetime!)
  - MN signals COA to the HA via the FA, HA acknowledges via FA to MN
  - these actions have to be secured by authentication
- Advertisement
  - HA advertises the IP address of the MN (as for fixed systems), i.e. standard routing information
  - routers adjust their entries, these are stable for a longer time (HA responsible for a MN over a longer period of time)
  - packets to the MN are sent to the HA,
  - independent of changes in COA/FA

Agent advertisement

0 7	8 15	16 23	24 31
type	code	checksum	
#addresses	addr. size	lifetime	
router address 1			
preference level 1			
router address 2			
preference level 2			

. . .

type	length	sequence number	
registration lifetime		R B H F M G V	reserved
COA 1			
COA 2			

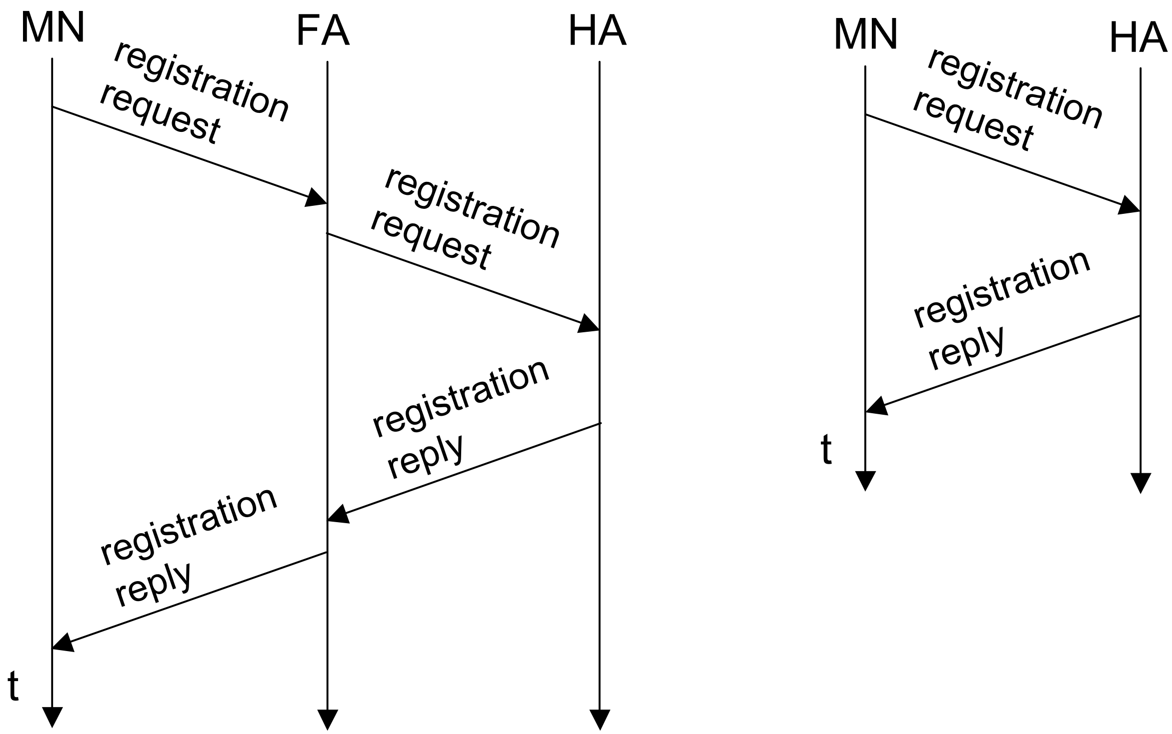
. . .

R: registration required
B: busy
H: home agent
F: foreign agent
M: minimal encapsulation
G: generic encapsulation
V: header compression

ICMP-Type = 0; Code = 0/16; Extension Type = 16

TTL = 1 Dest-Adr = 224.0.0.1 (multicast on link) or 255.255.255.255 (broadcast)

Registration



Goal: inform the home agent of current location of MN (COA-FA or co-located COA)

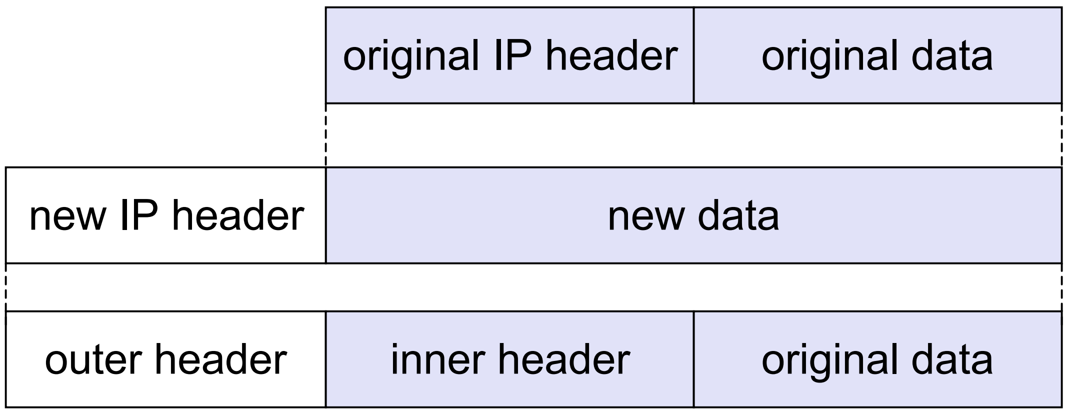
Registration expires automatically (lifetime)
Uses UDP port 434

Mobile IP registration request

0 7	8 15	16 23	24 31
type	S B D M G V rsv	lifetime	
home address			
home agent			
COA			
identification			
extensions . . .			

UDP packet on port 343
Type = 1 for registration request
S: retain prior mobility bindings
B: forward broadcast packets
D: co-located address=> MN decapsulates packets

Encapsulation



Encapsulation I

- Encapsulation of one packet into another as payload
  - e.g. IPv6 in IPv4 (6Bone), Multicast in Unicast (Mbone)
  - here: e.g. IP-in-IP-encapsulation, minimal encapsulation or GRE (Generic Record Encapsulation)
- IP-in-IP-encapsulation (mandatory in RFC 2003)
  - tunnel between HA and COA

<table>
<tr><td>ver.</td><td>IHL</td><td>TOS</td><td colspan="2">length</td></tr>
<tr><td colspan="3">IP identification</td><td>flags</td><td>fragment offset</td></tr>
<tr><td colspan="2">TTL</td><td>IP-in-IP</td><td colspan="2">IP checksum</td></tr>
<tr><td colspan="5">IP address of HA</td></tr>
<tr><td colspan="5">Care-of address COA</td></tr>
<tr><td>ver.</td><td>IHL</td><td>TOS</td><td colspan="2">length</td></tr>
<tr><td colspan="3">IP identification</td><td>flags</td><td>fragment offset</td></tr>
<tr><td colspan="2">TTL</td><td>lay. 4 prot.</td><td colspan="2">IP checksum</td></tr>
<tr><td colspan="5">IP address of CN</td></tr>
<tr><td colspan="5">IP address of MN</td></tr>
<tr><td colspan="5">TCP/UDP/ ... payload</td></tr>
</table>

Encapsulation II

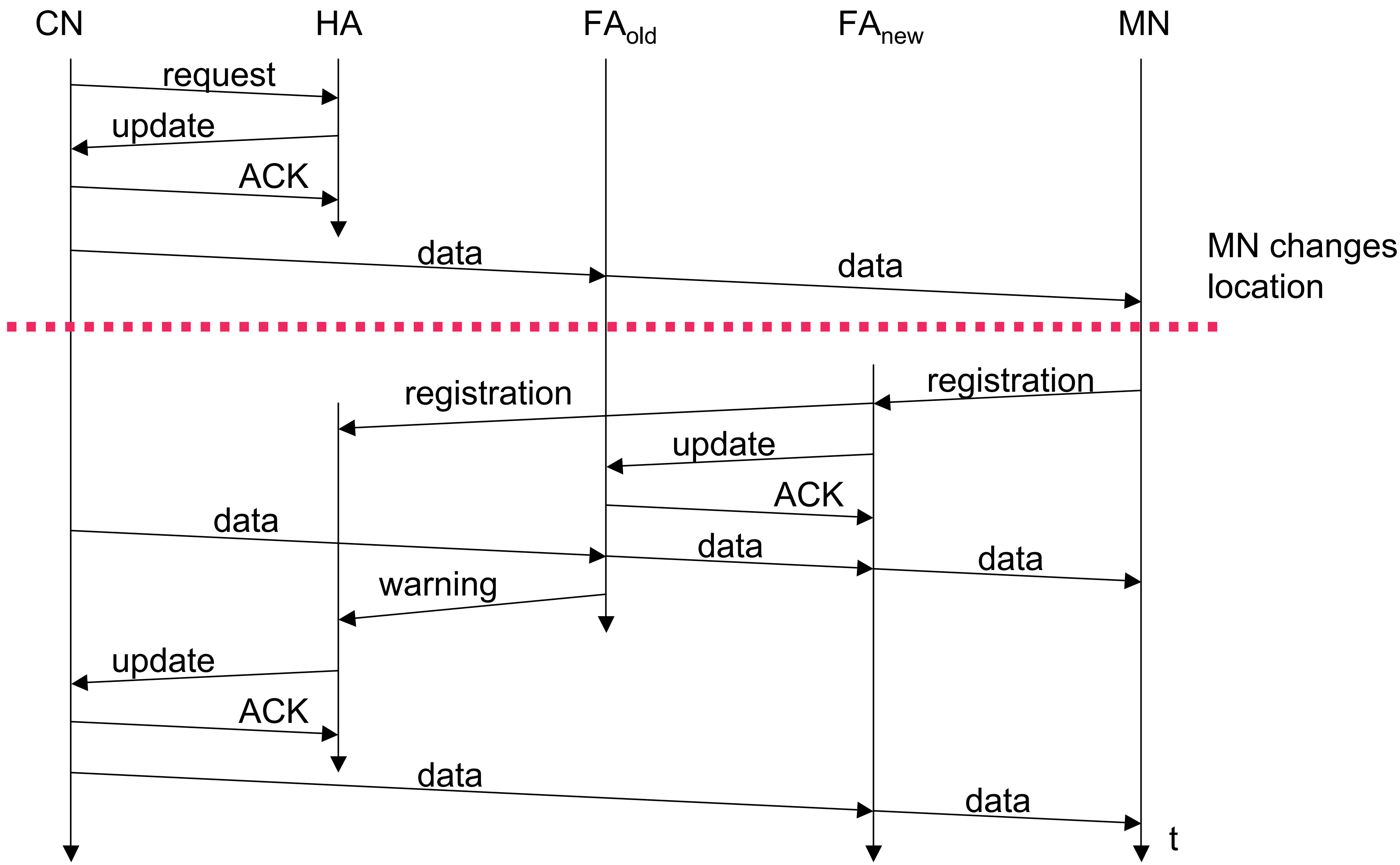
- Minimal encapsulation (optional) [RFC2004]
  - avoids repetition of identical fields
  - e.g. TTL, IHL, version, TOS
  - only applicable for unfragmented packets, no space left for fragment identification

<table>
<tr><td>ver.</td><td>IHL</td><td colspan="2">TOS</td><td colspan="2">length</td></tr>
<tr><td colspan="4">IP identification</td><td>flags</td><td>fragment offset</td></tr>
<tr><td colspan="2">TTL</td><td colspan="2">min. encap.</td><td colspan="2">IP checksum</td></tr>
<tr><td colspan="6">IP address of HA</td></tr>
<tr><td colspan="6">care-of address COA</td></tr>
<tr><td colspan="2">lay. 4 protoc.</td><td>S</td><td>reserved</td><td colspan="2">IP checksum</td></tr>
<tr><td colspan="6">IP address of MN</td></tr>
<tr><td colspan="6">original sender IP address (if S=1)</td></tr>
<tr><td colspan="6">TCP/UDP/ ... payload</td></tr>
</table>

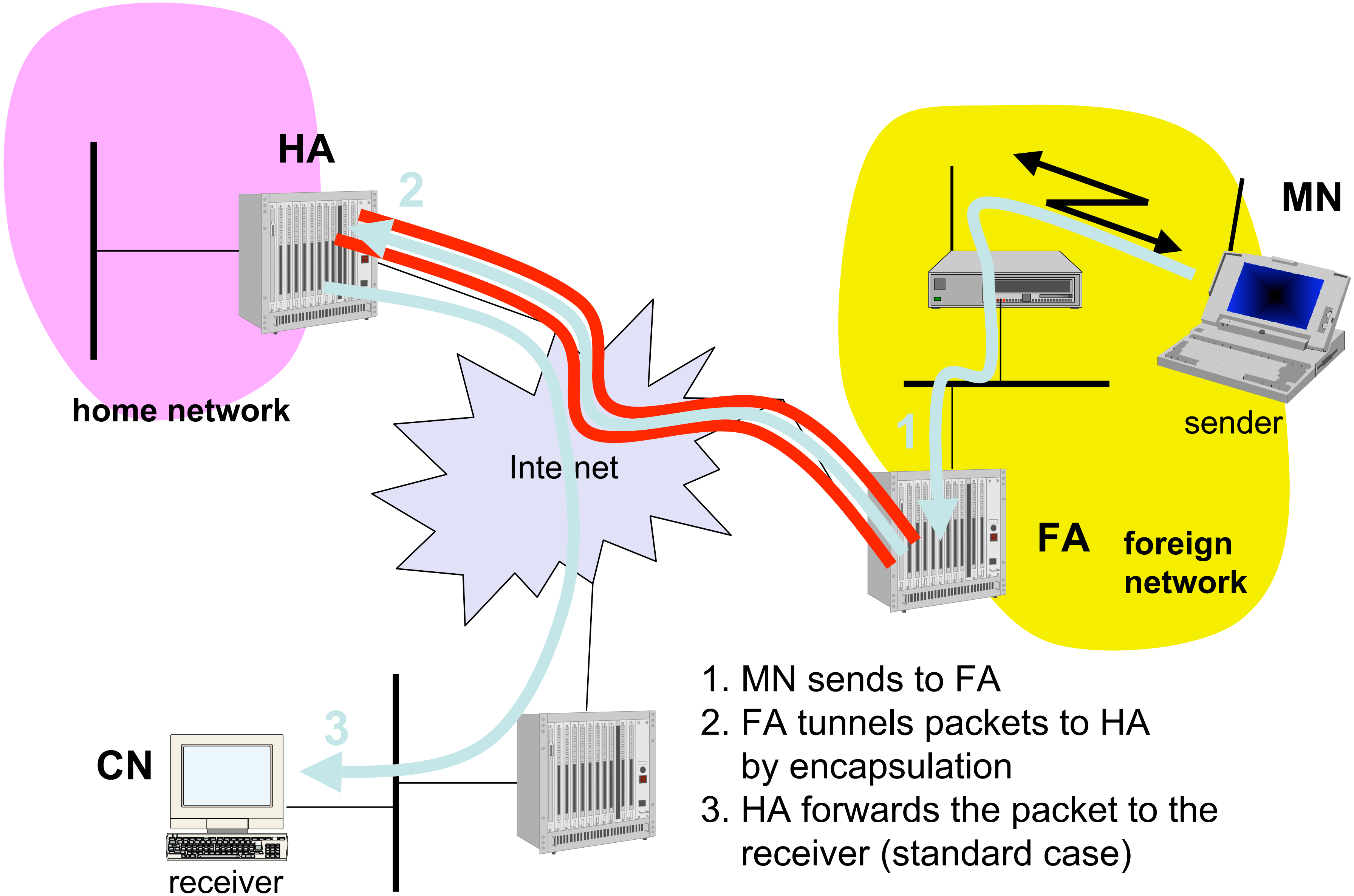
Optimization of packet forwarding

- Triangular Routing
  - sender sends all packets via HA to MN
  - higher latency and network load
- "Solutions"
  - sender learns the current location of MN
  - direct tunneling to this location
  - HA informs a sender about the location of MN
  - big security problems!
- Change of FA
  - packets on-the-fly during the change can be lost
  - new FA informs old FA to avoid packet loss, old FA now forwards remaining packets to new FA
  - this information also enables the old FA to release resources for the MN

Change of foreign agent



Reverse tunneling (RFC 2344)



1. MN sends to FA
2. FA tunnels packets to HA
   by encapsulation
3. HA forwards the packet to the
   receiver (standard case)

Mobile IP with reverse tunneling

- Routers accept often only “topological correct“ addresses (firewall)
  - a packet from the MN encapsulated by the FA is now topological correct
  - furthermore multicast and TTL problems solved (TTL in the home network correct, but MN is to far away from the receiver)
- Reverse tunneling does not solve
  - problems with *firewalls*, the reverse tunnel can be abused to circumvent security mechanisms (tunnel hijacking)
  - optimization of data paths, i.e. packets will be forwarded through the tunnel via the HA to a sender (double triangular routing)
- The new standard is backwards compatible
  - the extensions can be implemented easily and cooperate with current implementations without these extensions

Mobile IP and IPv6

- security is integrated and not an add-on, authentication of registration is included
- COA can be assigned via auto-configuration (DHCPv6 is one candidate), every node has address autoconfiguration
- no need for a separate FA, **all** routers perform router advertisement which can be used instead of the special agent advertisement
- MN can signal a sender directly the COA, sending via HA not needed in this case (automatic path optimization)
- “soft” hand-over, i.e. without packet loss, between two subnets is supported
    - MN sends the new COA to its old router
    - the old router encapsulates all incoming packets for the MN and forwards them to the new COA
    - authentication is always granted

Problems with Mobile IP

- Security
  - authentication with FA problematic, for the FA typically belongs to another organization
  - no protocol for key management and key distribution has been standardized in the Internet
  - patent and export restrictions
- Firewalls
  - typically mobile IP cannot be used together with firewalls, special set-ups are needed (such as reverse tunneling)
- QoS
  - many new reservations in case of RSVP
  - tunneling makes it hard to give a flow of packets a special treatment needed for the QoS
- Security, firewalls, QoS etc. are topics of current research and discussions!